

# $K_{\text{frak}}$ , Unrolling and the $\tilde{T} \cdot m$ Duality

Why the Fractal Correction Is Not a Property of Time Unrolling

Doc. 333

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## Contents

### Preliminary note

In the FFGFT corpus, “unrolling” has hitherto always meant **type-I decompactification**: the measurement act selects the mass circle as time,  $S_m^1 \rightarrow \mathbb{R}_t$ , discarding the winding number — a lossy step (Doc. 285, Doc. 248, Doc. 270). Doc. 332 introduces alongside this a different operation: the **canonical function pullback**  $\Phi = p^* : L^2(S_m^1) \rightarrow \text{Per}_{R_m}(\mathbb{R}_t)$ , which is lossless and exact — a pure change of coordinates with no loss of information.

This document places both operations side by side, clarifies the role of  $K_{\text{frak}}$  in each case, and shows that the recursion follows necessarily from the  $\tilde{T} \cdot m$  duality.

### .1 Two Meanings of “Unrolling” in the Corpus

#### Type I: the measurement act (lossy)

The meaning used so far in the corpus (Doc. 285 § “How FFGFT unrolls”): the embedded observer selects the mass circle  $S_m^1$  as the time axis. The map

$$S_m^1 \longrightarrow \mathbb{R}_t, \quad \tilde{T} \mapsto t,$$

is **lossy**: the winding number is discarded, the discrete spectrum of the compact circle does not appear in  $\mathbb{R}_t$ . This is not a mathematical operation of the theory but the bridge to observation — the torus remains the fundamental structure,  $\mathbb{R}_t$  is the image (Doc. 285). At the same time the recursion  $\xi^n$  unrolls into a continuous dilation flow  $t \mapsto \xi t$  with log-period  $|\ln \xi| = \ln 7500 \approx 8.92$  (Doc. 285 § “The recursion unrolls along with it”).

#### Type III: the coordinate change (lossless)

The meaning introduced in Doc. 332: the canonical function pullback

$$\Phi = p^* : L^2(S_m^1) \longrightarrow \text{Per}_{R_m}(\mathbb{R}_t), \quad (\Phi f)(t) = f(t \bmod R_m),$$

is an **exact isometry** — no loss of information, no winding number discarded, Fourier basis and Parseval identity preserved. It is a change of representation within isomorphic Hilbert spaces (R96, Doc. 332). The lossy direction lies at  $\mathbb{R}_t \rightarrow S_m^1$ , not at the pullback itself.

|                   | Type I (measurement act)         | Type III (pullback)                               |
|-------------------|----------------------------------|---------------------------------------------------|
| Operation         | $S_m^1 \rightarrow \mathbb{R}_t$ | $L^2(S_m^1) \cong \text{Per}_{R_m}(\mathbb{R}_t)$ |
| Loss              | winding number discarded         | dis- none                                         |
| Character         | physical act                     | mathematical change of representation             |
| Winding number    | not preserved                    | preserved                                         |
| Information loss  | yes                              | no                                                |
| Corpus references | Doc. 285, 248, 270               | Doc. 332 (R96)                                    |

## .2 The Role of $K_{\text{frak}}$ in Both Cases

### At the type-III pullback: invisible

A Laplace operator on the compact  $T^4$  is local — it cannot carry a cumulative quantity (Doc. 314).  $K_{\text{frak}}$  therefore does not enter through the operator but through the **bridge constants**  $E_0$ ,  $\nu$  at the scale conversion (Doc. 314, R72).

In the rolled-up state on  $S_m^1$  the period  $R_m = \hbar/(mc^2)$  is exact in natural units — no  $K_{\text{frak}}$ . Only when  $m$  is substituted from the mass formula does  $R_m$  carry the correction:

$$m_i = r_i \cdot \xi_0^{p_i} \cdot \nu \cdot K_{\text{frak}} \quad \Rightarrow \quad R_m^{\text{corr.}} = \frac{\hbar}{m_i^{\text{bare}} \cdot K_{\text{frak}} \cdot c^2} = \frac{R_m^{\text{bare}}}{K_{\text{frak}}}.$$

$K_{\text{frak}}$  sits in the *choice of  $R_m$* , not in the pullback mechanism. The pullback itself is exact in both cases (bare or corrected).

### At the type-I measurement act: visible as a scale correction

At the lossy unrolling,  $K_{\text{frak}}$  appears as a correction factor of the bridge constants: the ratio of the two  $E_0$  values (bare and  $\alpha$ -anchored) is  $1/\sqrt{K_{\text{frak}}}$  (R72, Doc. 314). The correction belongs to the *scale structure* — the transition between the compact and the observable representation — not to the time geometry itself.

|                   | Type I                                                   | Type III                                              |
|-------------------|----------------------------------------------------------|-------------------------------------------------------|
| $K_{\text{frak}}$ | visible in bridge constants ( $E_0$ ratio, mass formula) | invisible at the isometry step; sits in $R_m$ via $m$ |
| Where exactly     | at the SI transition                                     | in the choice of period                               |

## .3 Mass and Time Unroll Together

In the FFGFT compactification,  $\tilde{T}$  and  $m$  are a **single geometric dimension**:

$$\tilde{T} \cdot m = 1 \quad (\text{fundamental relation, Doc. 285}).$$

They are not two independent quantities. At the type-III pullback, the pullback therefore unrolls  $m$  and  $t$  *together*: the entire mode  $f \in L^2(S_m^1)$  is unrolled and the ratio  $\tilde{T}/m$  remains constant. The physics does not change, only the coordinate representation.

Unrolling  $t$  alone — dropping  $R_m$  — would violate  $\tilde{T} \cdot m = 1$  and is structurally inadmissible.  $K_{\text{frak}}$  is *not* the price of unrolling, but the price of the **artificial separation** of  $m$  and  $t$  at the transition to absolute SI units.

## .4 The Recursion Is Necessarily Bound to the Duality

### Stable particle: Case B

For a stable, isolated particle  $\xi$  is frozen (Case B, Doc. 315, Doc. 313): the winding number is rational  $74/75$ , closure occurs after exactly 75 revolutions.  $K_{\text{frak}} = 74/75$  is the closed bookkeeping of this rolled-up geometry — exact in fractional arithmetic, because  $\xi$  is set rationally (Doc. 315).

### Dynamics: Case C and the recursion chain

As soon as dynamics enter — interaction, RG flow, mass change — a deeper consequence of  $\tilde{T} \cdot m = 1$  comes into play: if  $m$  changes,  $\tilde{T} = 1/m$  changes necessarily with it. The period  $R_m = \hbar/(mc^2)$  changes, the time scale of the physics changes, and this feeds back onto the effective mass. The recursion (Doc. 295, Doc. 146) reads:

$$\xi_{n+1} = \xi_n (1 - 100 \xi_n).$$

The closure deficit per revolution no longer remains constant but runs cumulatively. The rational 75-closure becomes a logarithmic spiral (Case C, Doc. 295):

$$\rho = 75 e^{\beta/2\pi} \quad (\text{equiangular spiral, self-similarity ratio } e).$$

In Case C,  $K_{\text{frak}}$  is no longer a constant but an effectively scale-dependent quantity. The multiplicative form  $(1 - \xi)^{100}$  is the natural bookkeeping of this running case; its distance from  $74/75$  — the binomial term  $4950 \xi^2$  (Doc. 315) — is precisely the difference between frozen and running recursion.

### Why Case A is excluded

Case A ( $K_{\text{frak}} = 1$ , closure after one revolution) would mean:  $\tilde{T} \cdot m = 1$  generates no recursion, because  $m$  never changes — a static, interaction-free universe with  $D_f = 3$  exactly. This contradicts the observed particle structure and the basic framework.

### Summary of the three cases

| Case | $\xi$               | $K_{\text{frak}}$ | Binding to duality                                                  |
|------|---------------------|-------------------|---------------------------------------------------------------------|
| A    | constant, $d = 0$   | $= 1$             | no feedback, excluded                                               |
| B    | frozen              | $= 74/75$ exact   | stable mode, duality at rest                                        |
| C    | running (recursion) | scale-dependent   | duality generates running feedback<br>$m \leftrightarrow \tilde{T}$ |

$K_{\text{frak}} = 74/75$  is the **projection of Case C onto Case B**: exact in the rolled-up, static description; first approximation in the dynamic one.

## Summary

*"Unrolling" in the FFGFT corpus has hitherto always meant the lossy measurement act (Type I, Doc. 285): winding number discarded,  $S_m^1 \rightarrow \mathbb{R}_t$ . The function pullback from Doc. 332 is a different, lossless operation (Type III): an exact change of representation that unrolls  $m$  and  $t$  together and at which  $K_{\text{frak}}$  plays no role.  $K_{\text{frak}}$  sits in the choice of the period  $R_m$  via the mass formula and becomes visible only at the transition to absolute SI values. The recursion  $\xi_{n+1} = \xi_n(1 - 100 \xi_n)$  is necessarily bound to  $\tilde{T} \cdot m = 1$ : every mass change entails a time change, which feeds back recursively onto the mass. Case B (stable particle,  $K_{\text{frak}} = 74/75$ ) is the approximation for frozen duality; Case C (running recursion, log-spiral) is the complete dynamic description."*

**References:** Doc. 133 ( $K_{\text{frak}}$  derivation), Doc. 146 (recursion), Doc. 248 (observer), Doc. 270 (type-I theorem), Doc. 285 (decompactification), Doc. 295 (log-spiral), Doc. 313 (three cases), Doc. 314 (local operator, bridge constants), Doc. 315 (form question  $K_{\text{frak}}$ ), Doc. 332 (pullback isometry, R96).